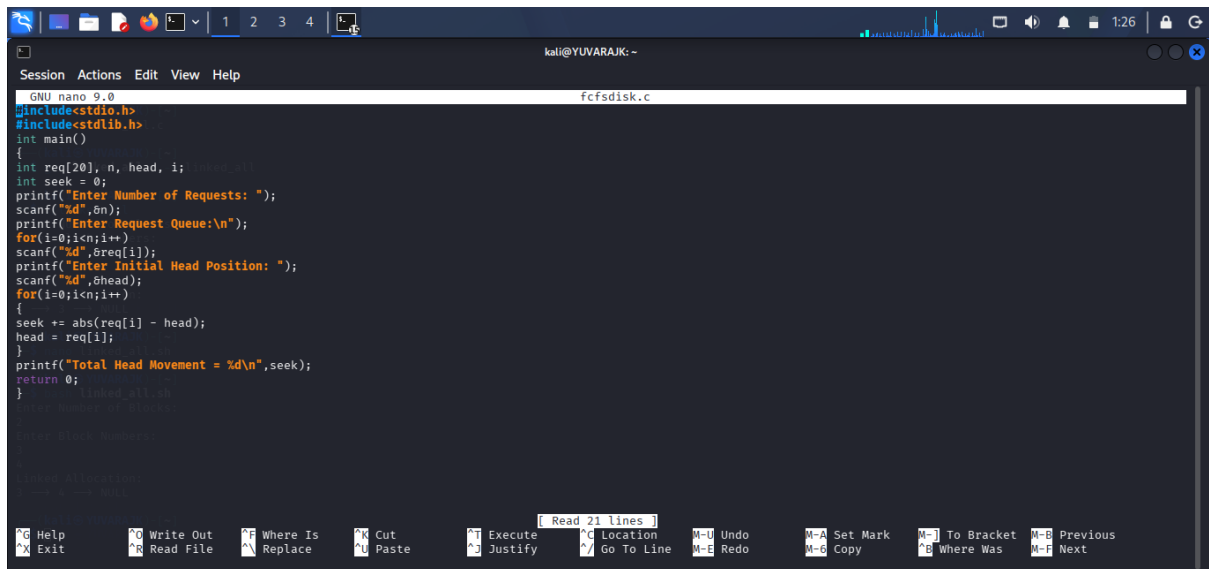
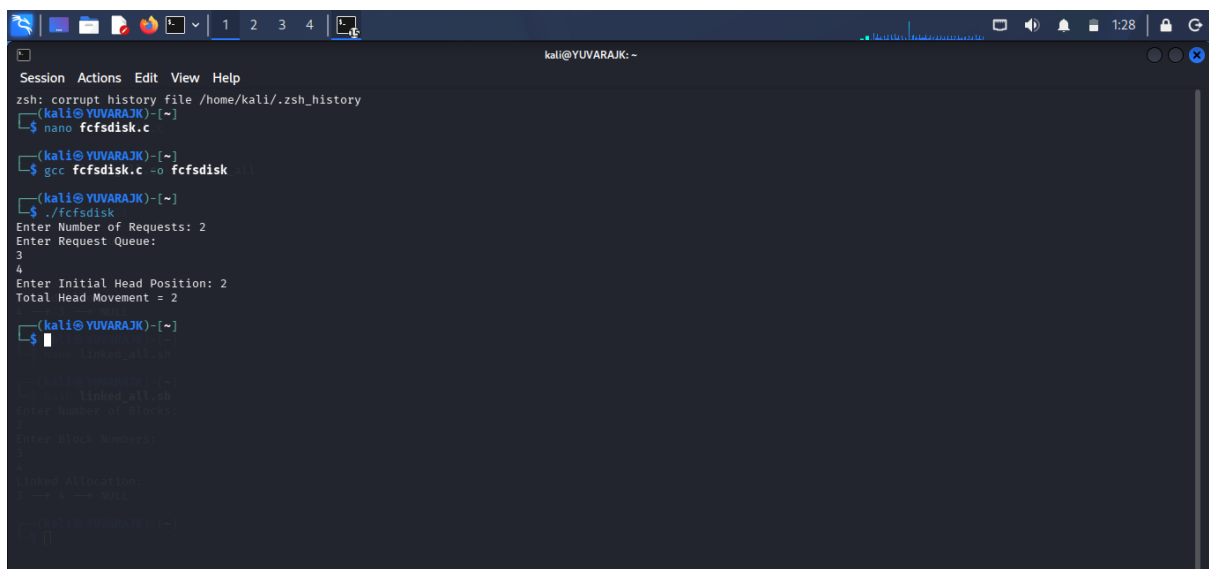


C PROGRAMMING :FCFS DISK



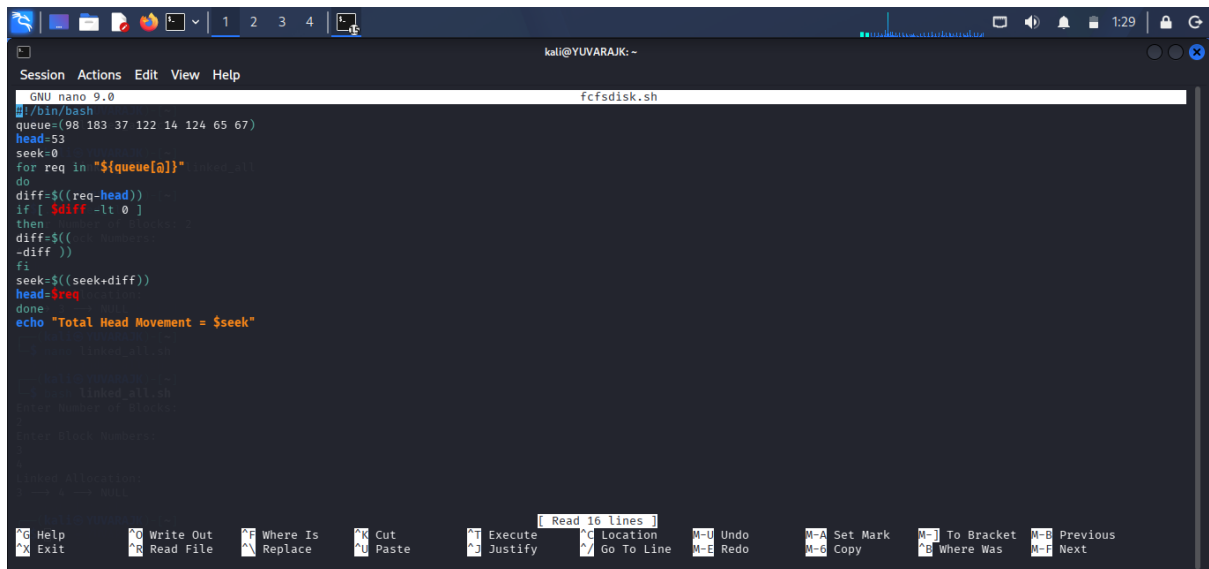
```
GNU nano 9.0 fcfsdisk.c
#include<stdio.h>
#include<stdlib.h>
int main()
{
    int req[20], n, head, i;
    int seek = 0;
    printf("Enter Number of Requests: ");
    scanf("%d", &n);
    printf("Enter Request Queue:\n");
    for(i=0; i<n; i++)
    {
        scanf("%d", &req[i]);
    }
    printf("Enter Initial Head Position: ");
    scanf("%d", &head);
    for(i=0; i<n; i++)
    {
        seek += abs(req[i] - head);
        head = req[i];
    }
    printf("Total Head Movement = %d\n", seek);
    return 0;
}
```

OUTPUT :



```
zsh: corrupt history file /home/kali/.zsh_history
(kali@YUVARAJK)-[~]
$ nano fcfsdisk.c
(kali@YUVARAJK)-[~]
$ gcc fcfsdisk.c -o fcfsdisk
(kali@YUVARAJK)-[~]
$ ./fcfsdisk
Enter Number of Requests: 2
Enter Request Queue:
3
4
Enter Initial Head Position: 2
Total Head Movement = 2
(kali@YUVARAJK)-[~]
$
```

SHELL PROGRAMMING : FCFS DISK

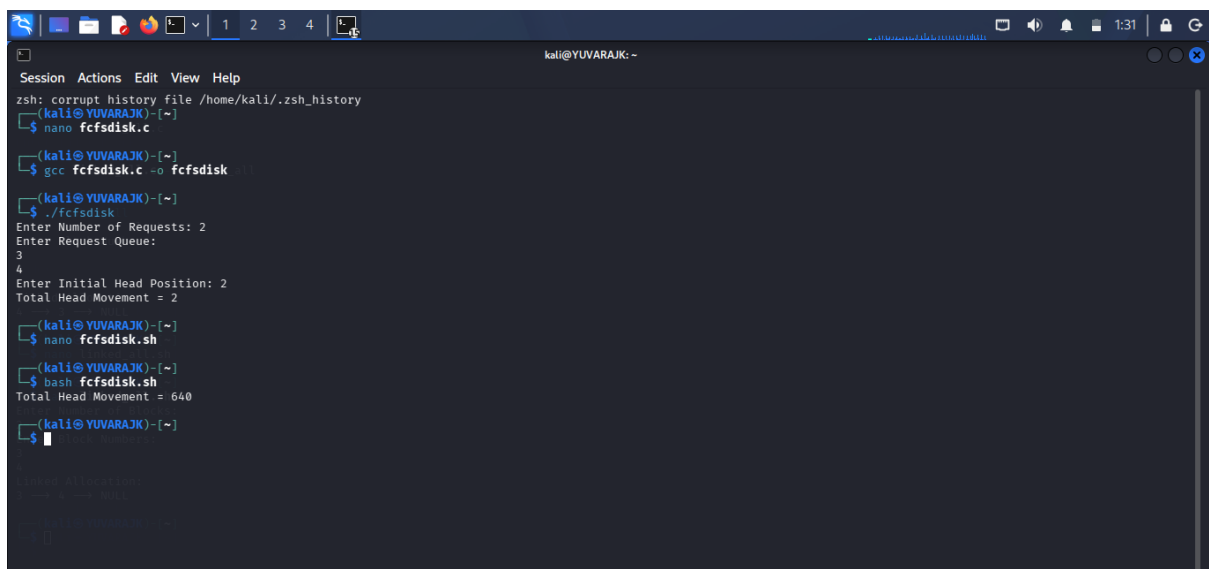


The screenshot shows a terminal window with the nano 9.0 editor open. The file being edited is fcfsdisk.sh. The script contains the following code:

```
#!/bin/bash
queue=(98 183 37 122 14 124 65 67)
head=53
seek=0
for req in "${queue[@]}"; do
    diff=$((req-head))
    if [ $diff -lt 0 ]; then
        then head=$((head-diff))
    fi
    seek=$((seek+diff))
    head=$req
done
echo "Total Head Movement = $seek"
```

The nano editor interface includes a menu bar with Session, Actions, Edit, View, and Help. The status bar at the bottom shows various keyboard shortcuts for editing and navigation.

OUTPUT :

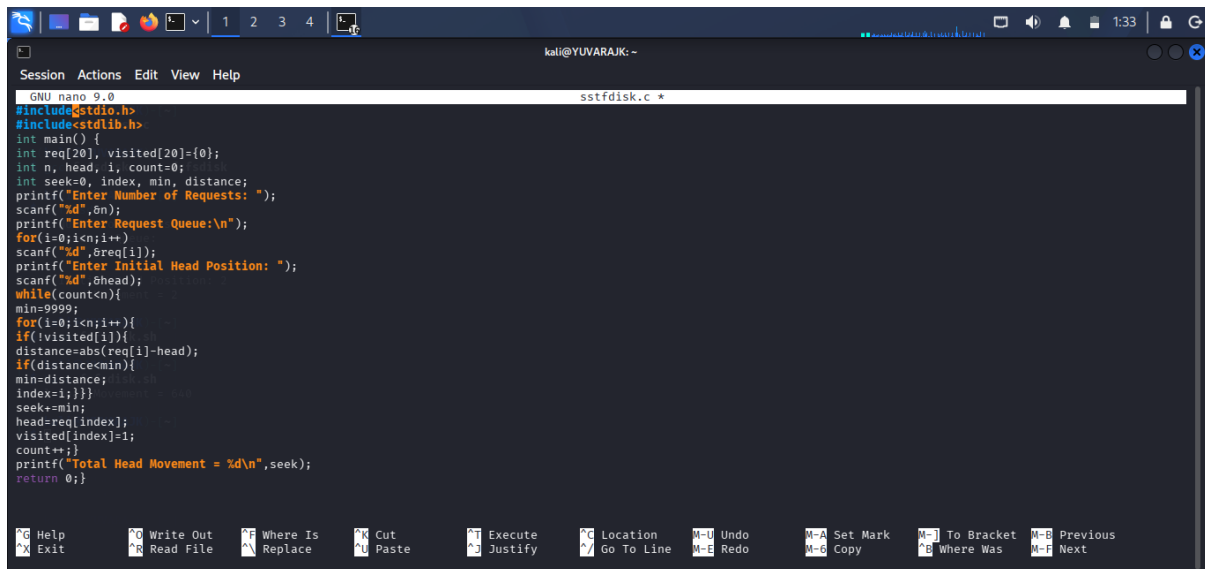


The screenshot shows a terminal window with the following commands and output:

```
zsh: corrupt history file /home/kali/.zsh_history
(kali@YUVARAJK)-[~]
$ nano fcfsdisk.c
(kali@YUVARAJK)-[~]
$ gcc fcfsdisk.c -o fcfsdisk
(kali@YUVARAJK)-[~]
$ ./fcfsdisk
Enter Number of Requests: 2
Enter Request Queue:
3
4
Enter Initial Head Position: 2
Total Head Movement = 2
(kali@YUVARAJK)-[~]
$ nano fcfsdisk.sh
(kali@YUVARAJK)-[~]
$ bash fcfsdisk.sh
Total Head Movement = 640
(kali@YUVARAJK)-[~]
$
```

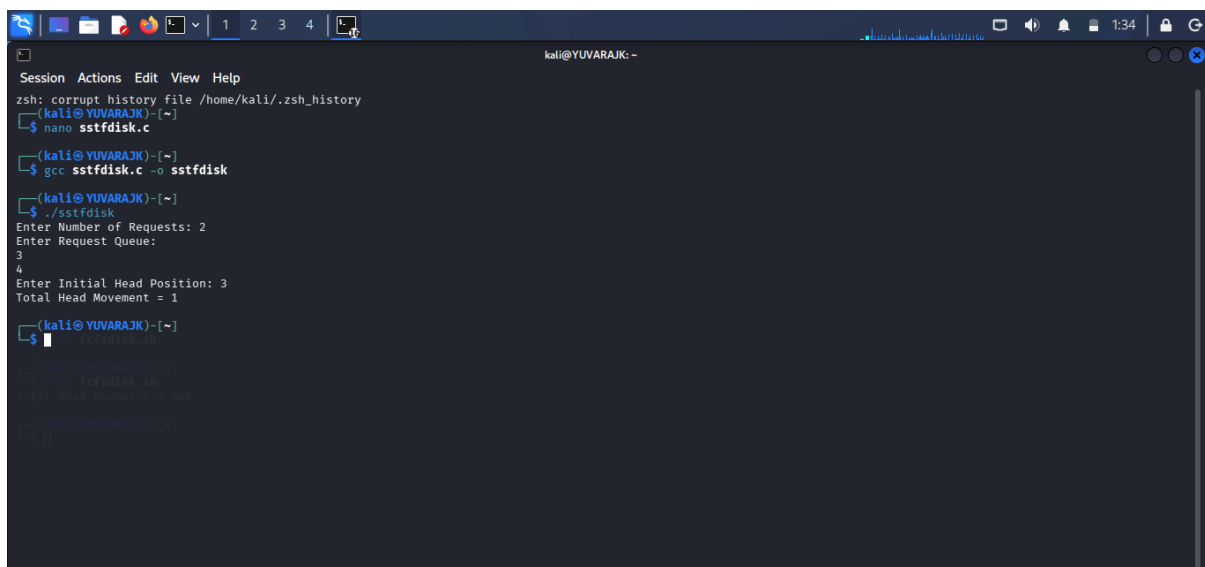
The terminal shows the user editing fcfsdisk.c, compiling it to fcfsdisk, and then running it. The script prompts for the number of requests (2) and the request queue (3, 4). It also prompts for the initial head position (2) and outputs the total head movement (2). The user then edits fcfsdisk.sh and runs it, which outputs a total head movement of 640.

C PROGRAMMING : SSTF DISK



```
GNU nano 9.0 sstfdisk.c *
#include<stdio.h>
#include<stdlib.h>
int main() {
    int req[20], visited[20]={0};
    int n, head, i, count=0;
    int seek=0, index, min, distance;
    printf("Enter Number of Requests: ");
    scanf("%d",&n);
    printf("Enter Request Queue:\n");
    for(i=0;i<n;i++){
        scanf("%d",&req[i]);
    }
    printf("Enter Initial Head Position: ");
    scanf("%d",&head);
    while(count<n){
        min=9999;
        for(i=0;i<n;i++){
            if(!visited[i]){
                distance=abs(req[i]-head);
                if(distance<min){
                    min=distance;
                    index=i;
                }
            }
        }
        head=req[index];
        visited[index]=1;
        count++;
        printf("Total Head Movement = %d\n",seek);
    }
    return 0;
}
```

OUTPUT :



```
zsh: corrupt history file /home/kali/.zsh_history
(kali@YUVARAJK)-[~]
$ nano sstfdisk.c
(kali@YUVARAJK)-[~]
$ gcc sstfdisk.c -o sstfdisk
(kali@YUVARAJK)-[~]
$ ./sstfdisk
Enter Number of Requests: 2
Enter Request Queue:
3
4
Enter Initial Head Position: 3
Total Head Movement = 1
(kali@YUVARAJK)-[~]
$
```

SHELL PROGRAMMING : SSTF DISK

The screenshot shows a Kali Linux terminal window. The title bar indicates the user is 'kali@YUVARAJIK' in the home directory. The terminal is running a nano editor session for a file named 'sstfdisk.sh'. The script's output is as follows:

```
GNU nano 9.0 sstfdisk.sh
#!/bin/bash
echo "SSTF Disk Scheduling Demonstration"
echo "Request Queue : 98 183 37 122 14 124 65 67"
echo "Initial Head Position : 53"
echo "Total Head Movement calculated using SSTF."
```

The bottom of the terminal features a command palette with the following options:

- Help
- Write Out
- Where Is
- Cut
- Execute
- Justify
- Location
- Go To Line
- Undo
- Redo
- Set Mark
- Copy
- To Bracket
- Where Was
- Previous
- Next

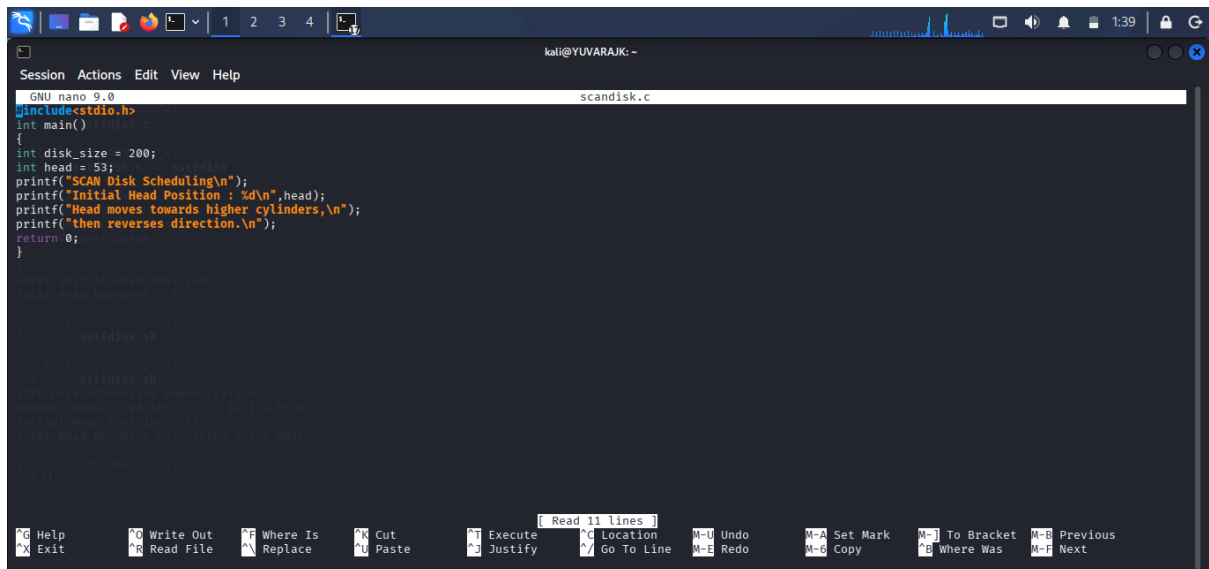
The 'Read 5 lines' option is currently selected in the palette.

OUTPUT :

```

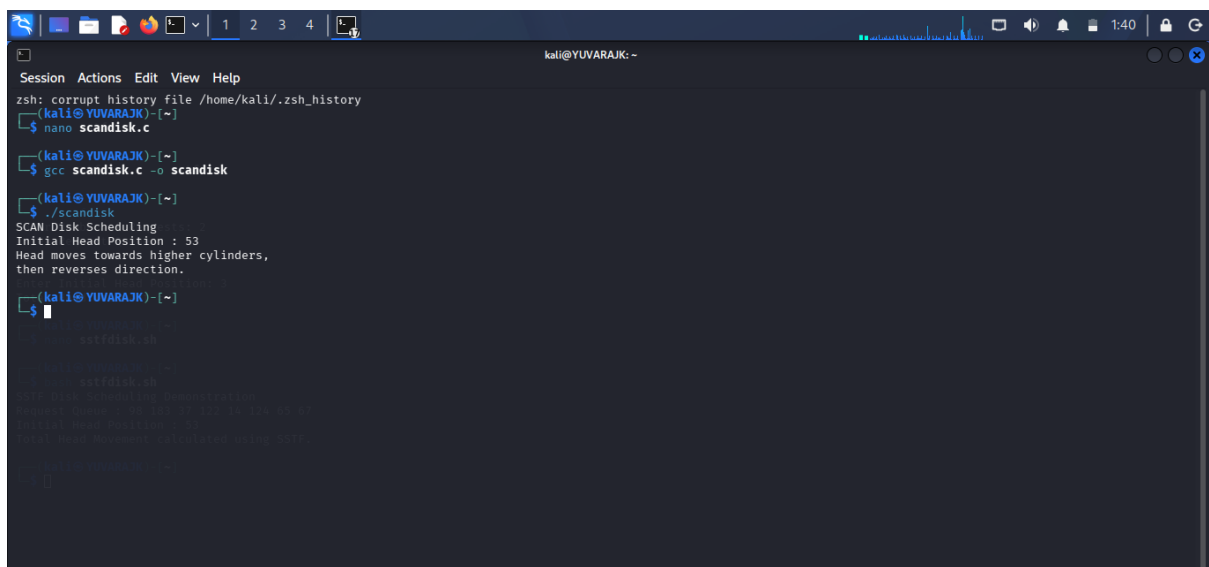
kali@YUVARAJK: ~
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
kali@YUVARAJK: ~
$ nano sstfdisk.c
kali@YUVARAJK: ~
$ gcc sstfdisk.c -o sstfdisk
kali@YUVARAJK: ~
$ ./sstfdisk
Enter Number of Requests: 2
Enter Request Queue:
3
4
Enter Initial Head Position: 3
Total Head Movement = 1
kali@YUVARAJK: ~
$ nano sstfdisk.sh
kali@YUVARAJK: ~
$ bash sstfdisk.sh
SSTF Disk Scheduling Demonstration
Request Queue : 98 183 37 122 14 124 65 67
Initial Head Position : 53
Total Head Movement calculated using SSTF.
kali@YUVARAJK: ~
$
```

C PROGRAMMING : SCAN DISK



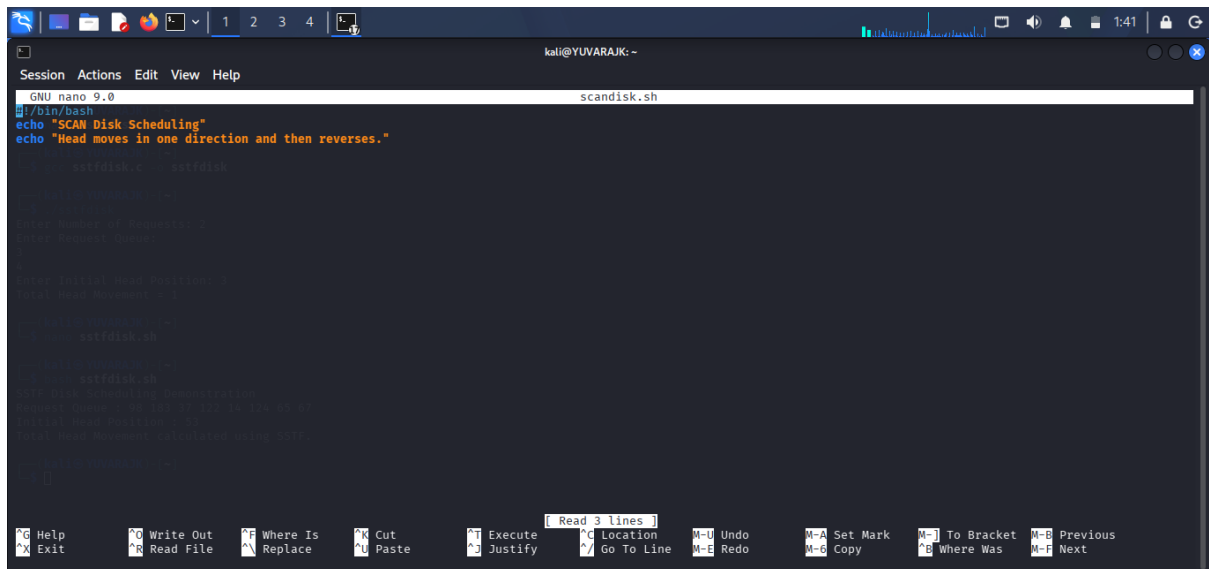
```
GNU nano 9.0 scandisk.c
#include<stdio.h>
int main()
{
    int disk_size = 200;
    int head = 53;
    printf("SCAN Disk Scheduling\n");
    printf("Initial Head Position : %d\n",head);
    printf("Head moves towards higher cylinders,\n");
    printf("then reverses direction.\n");
    return 0;
}
```

OUTPUT :



```
zsh: corrupt history file /home/kali/.zsh_history
(kali@YUVARAJK)-[~]
$ nano scandisk.c
(kali@YUVARAJK)-[~]
$ gcc scandisk.c -o scandisk
(kali@YUVARAJK)-[~]
$ ./scandisk
SCAN Disk Scheduling
Initial Head Position : 53
Head moves towards higher cylinders,
then reverses direction.
(kali@YUVARAJK)-[~]
$
```

SHELL PROGRAMMING : SCAN DISK



The screenshot shows a terminal window with the nano 9.0 editor open. The file being edited is named 'scandisk.sh'. The script content is as follows:

```
#!/bin/bash
echo "SCAN Disk Scheduling"
echo "Head moves in one direction and then reverses."

# Initial Head Position
# Initial Head Position : 53
# Head moves towards higher cylinders,
# then reverses direction.

# Initial Head Position : 53
# Head moves towards higher cylinders,
# then reverses direction.

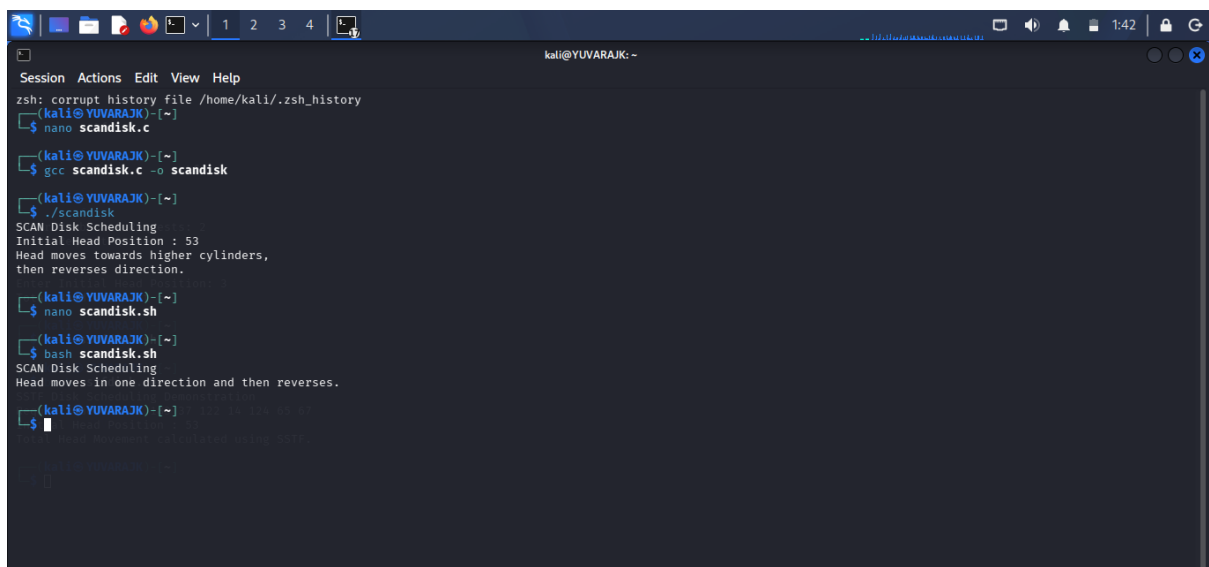
# Initial Head Position : 53
# Head moves towards higher cylinders,
# then reverses direction.

# Initial Head Position : 53
# Head moves towards higher cylinders,
# then reverses direction.

# Initial Head Position : 53
# Head moves towards higher cylinders,
# then reverses direction.

# Initial Head Position : 53
# Head moves towards higher cylinders,
# then reverses direction.
```

OUTPUT :



The screenshot shows a terminal window with the following commands and output:

```
zsh: corrupt history file /home/kali/.zsh_history
(kali@YUVARAJK)-[~]
$ nano scandisk.c
(kali@YUVARAJK)-[~]
$ gcc scandisk.c -o scandisk
(kali@YUVARAJK)-[~]
$ ./scandisk
SCAN Disk Scheduling
Initial Head Position : 53
Head moves towards higher cylinders,
then reverses direction.
(kali@YUVARAJK)-[~]
$ nano scandisk.sh
(kali@YUVARAJK)-[~]
$ bash scandisk.sh
SCAN Disk Scheduling
Head moves in one direction and then reverses.
(kali@YUVARAJK)-[~]
$
```

C PROGRAMMING : CIRCULAR SCAN

The screenshot shows a terminal window with the nano text editor open, editing a file named `circularscan.c`. The code implements the C-SCAN disk scheduling algorithm. It starts with including `stdio.h` and a `main` function. Inside `main`, it initializes `head` to 53 and prints the initial head position. It then enters a loop where it prints the head position and moves it in one direction. The code is as follows:

```

GNU nano 9.0 circularscan.c
#include<stdio.h>
int main()
{
    int head = 53;
    printf("C-SCAN Disk Scheduling\n");
    printf("Initial Head Position : %d\n",head);
    printf("Head moves in one direction.\n");
    printf("After reaching the end, it returns to the beginning.\n");
    return 0;
}

```

The terminal window has a dark background with a light blue border. The top bar shows the session name `kali@YUVARAJK-`. The bottom status bar indicates that 10 lines were written. The bottom of the terminal shows various keyboard shortcuts for nano, such as `Ctrl+H` for Help, `Ctrl+W` for Write Out, etc.

OUTPUT :

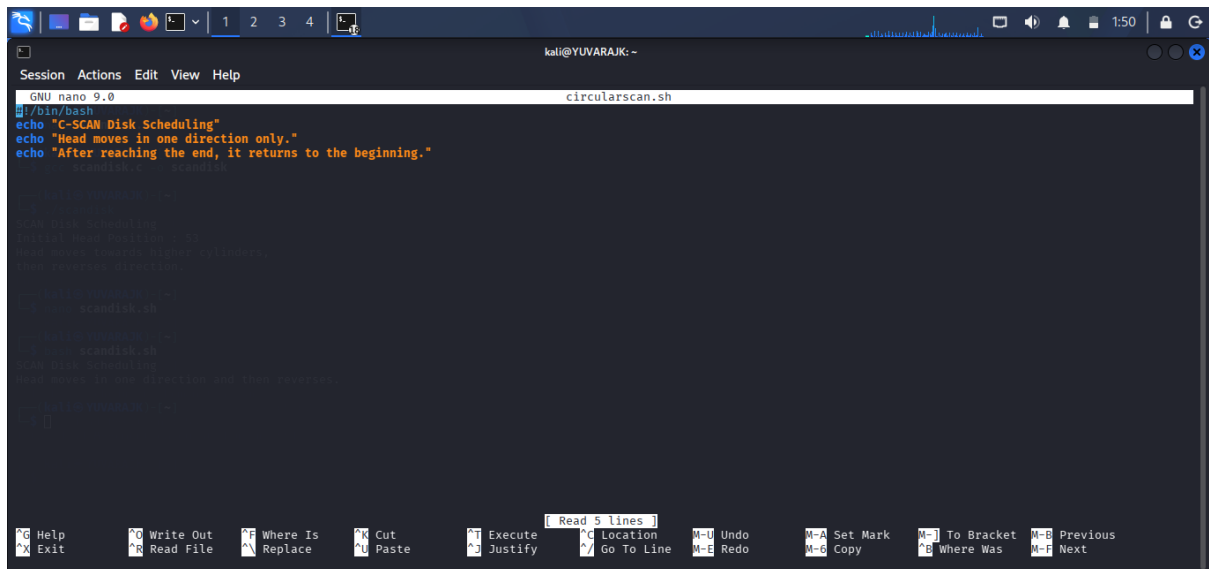
The screenshot shows a Kali Linux terminal window with the following content:

```

kali@YUVARAJK: ~
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
(kali@YUVARAJK)-[~]
$ nano dscandisk.c
(kali@YUVARAJK)-[~]
$ gcc circularscan.c -o circularscan
(kali@YUVARAJK)-[~]
$ ./circularscan
C-SCAN Disk Scheduling
Initial Head Position : 53
Head moves in one direction.
After reaching the end, it returns to the beginning.
(kali@YUVARAJK)-[~]
$

```

SHELL PROGRAMMING : CIRCULAR SCAN



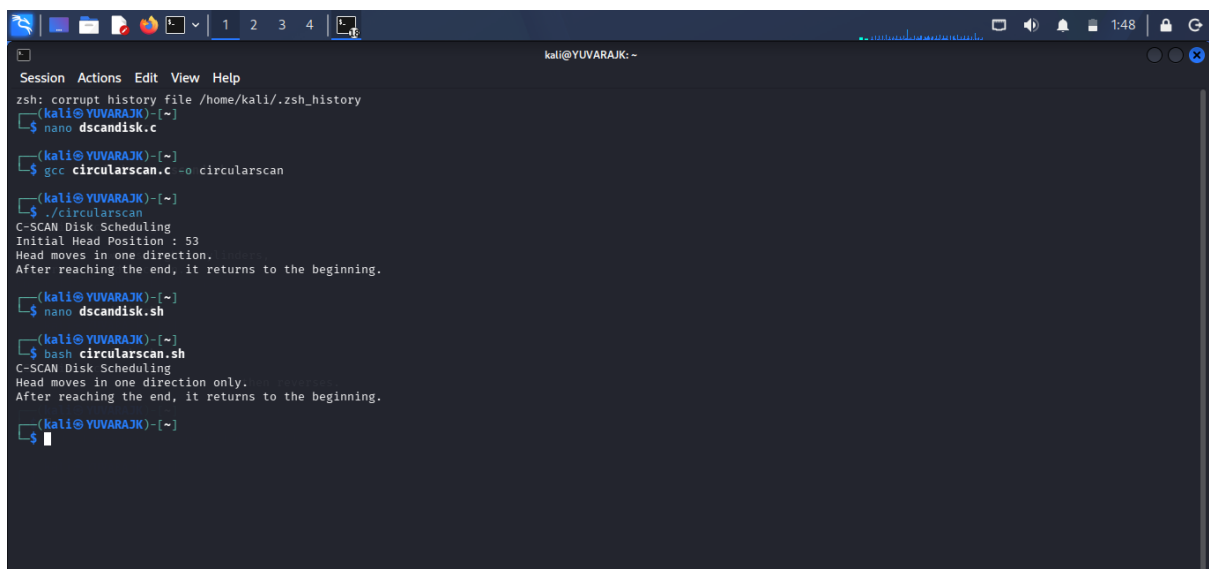
The screenshot shows a terminal window with the nano 9.0 editor open. The file being edited is circularscan.sh. The script content is as follows:

```
#!/bin/bash
echo "C-SCAN Disk Scheduling"
echo "Head moves in one direction only."
echo "After reaching the end, it returns to the beginning."

# Circular Scan
# C-SCAN Disk Scheduling
# Initial Head Position : 53
# Head moves towards higher cylinders.
# And reverse direction.

# Circular Scan
# C-SCAN Disk Scheduling
# Initial Head Position : 53
# Head moves in one direction and then reverse.
```

OUTPUT :



The screenshot shows a terminal window with the following commands and output:

```
zsh: corrupt history file /home/kali/.zsh_history
(kali@YUVARAJK)-[~]
$ nano dscandisk.c
(kali@YUVARAJK)-[~]
$ gcc circularscan.c -o circularscan
(kali@YUVARAJK)-[~]
$ ./circularscan
C-SCAN Disk Scheduling
Initial Head Position : 53
Head moves in one direction.
After reaching the end, it returns to the beginning.
(kali@YUVARAJK)-[~]
$ nano dscandisk.sh
(kali@YUVARAJK)-[~]
$ bash circularscan.sh
C-SCAN Disk Scheduling
Head moves in one direction only.
After reaching the end, it returns to the beginning.
(kali@YUVARAJK)-[~]
$
```